OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
TECHNOLOGICAL UNIVERSITY DUBLIN

Name: _____

Student #: _____

Professional Penetration Testing

Practical Exam

Network

You have been given your own IP address range for running your penetration test. This is available on Brightspace.

- **The target network for the this exam is 10.10.10.0/24.**
- **Any IP above (and including) 10.10.10.150 on this network is OUT OF BOUNDS so do not scan or attack 10.10.10.150 or above.**
- **Also, any IP address below (and including) 10.10.10.100 is OUT OF BOUNDS so do not scan or attack 10.10.10.100 or below.**
- **You are only permitted to scan targets from the target network. Anyone found scanning, attacking or otherwise any part of the infrastructure will receive a score of zero for the practical exam, even if it's accidental!!!**
- **For the Wi-Fi part of the penetration test de-authenticating and packet injection are strictly not allowed. You will lose all wireless marks if you attempt to do this!!!**

Anyone found scanning, attacking or otherwise any part of the infrastructure will receive a score of zero, even if it's accidental.

(On a real live network, you often have sections you **CANNOT** touch… this is the same). If you think any part of the network has crashed, let me know ASAP and I'll reset the Images.

Target network: 10.10.10.0/24 – NOTE THAT CERTAIN IP ADDRESS ARE OUT OF SCOPE.

You have been given a range of IP address that you can allocate to your own machines. Your IP address can be found on Brightspace.

Verifying Answers: For each question you will need to show your answer and solution to your instructor who will sign off each question. Remember you'll only get graded for what you can verify and show to the instructor.

Testing connection to the network: There are 3 servers on the live network (you may also get the hosting OS. If you get any Windows 11 machines that's the host, so you can ignore it.)

Test Target: 10.10.10.10 you can try pinging this.. or nmap -sP 210.10.10.10. If you get a response, you are on the network, **good luck**.

Recreated from photographs of the sheets of the practical exam (18 May 2026). The name and student number lines are left blank. On the next pages the answers written on the sheets are highlighted, and the boxes show the marks written by the instructor.

11.5

Reconnaissance (15%)

For each of the questions listed below, you should show your instructor the tool/command used to find your answer.

Question 1

a. The eshopworld.com domain wasn't always owned by the same company. What was listed as the source for delivery delays on the older eshopworld website in Oct 2006?

(3 marks)

1

Tool Used: web.archive.org

Answer: https://web.archive.org/web/20061030075047/

http://www.eshopworld.com/

I didn't find the route, but only the page.

b. Hackers don't follow the rules, and they love going places that they are not meant to. Can you find the url of the admin login page that eShopWorld is trying to hide from the search bots?

(3 marks)

3

Tool Used: Robots.txt

Answer: /wb-admin/

c. What is the exact OS and Web server currently used to host the www.rte.ie website?

(3 marks)

1.5

Tool Used: whatweb

Answers: Web Server: Cloudflare

OS: — (answer: hidden by Cloudflare server)

OS question was not answered.

d. Your lecturer Phillip Fitzpatrtick has a public profile on Facebook and X. What is his Facebook name (handle) and X handle.

(3 marks)

3

Tool Used: google dorking

Answers: [removed from this copy]

e. Identify the IP address, hosting provider, and DNS records of www.tudublin.ie using only passive techniques.

(3 marks)

3

Tool Used: nslookup shodan.io

Answers: 166.117.192.233 Amazon

166.117.104.145 Cloud Provider

Recreated from photographs. Highlighted: answers written on the sheet (1c reworded; in 1d the lecturer's social media handles are left out). Light blue: a note added afterwards. Boxes: marks written by the instructor; circled: total of the page.

15

Wireless (15%)

For each of the questions listed below, you should show your instructor the tool/command used to find your answer.

Question 2

a. What is the security protocol used by the WIFI network with the ESSID "Pentest Exam" and determine either the key or passphrase accordingly.

(8 marks)

8

Tool Used: airodump-ng aircrack-ng

WEP

Key / Passphrase: F2:C7:BB:35:B9

b. What is the security protocol used by WIFI network from the wireless handshake capture wireless.cap (available on Brightspace) and determine the key or passphrase accordingly.

(7 marks)

7

Tool Used: aircrack-ng with Rockyou.txt file

WPA

Key / Passphrase: hammertime

Recreated from photographs. Highlighted: the answers written on the sheet. Boxes: the marks written by the instructor; the circled 15 is the total written at the top of the page.

Scanning (15%)

For each of the questions listed below, you should show your instructor the tool/command used to find your answer.

Question 3

a. What is the IP address of each active machine in scope on the target network? **(3 marks)**

3

IP Address 1: 10.10.10.101

IP Address 2: 10.10.10.102

IP Address 3: 10.10.10.109

b. What is the exact operating system is running on each of the active machines? **(3 marks)**

3

Machine 1: Linux 14.04

Machine 2: Linux 16.04

Machine 3: Windows XP SP1

c. List all open ports and detailed service information for all open ports on each machine. (TCP and UDP). Show and explain to your instructor **(4 marks)**

2

Tool Used: nmap -sV 10.10.10.101,102,109

I should have scanned all the ports with -p- (nmap -sV -p- 10.10.10.101,102,109).

d. List ALL possible vulnerabilities (detected by your vulnerability scanner) available on each target machine. Show and explain this to your instructor. **(4 marks)**

4

Tool Used: nessus and openvas

Recreated from photographs. Highlighted: answers written on the sheet (the answers to 3c and 3d were shown to the instructor and are added afterwards). Light blue: a note added afterwards. Boxes: marks written by the instructor; circled: total of the page.

Access (55%)

For each of the questions listed below, you should show your instructor the tool/command used to find your answer.

Question 4

a. Gain root access to machine 1 **(15 marks)**

15

Tool(s) Used: ssh vagrant@10.10.10.101

(Password 'vagrant')

Exploit(s) Used: sudo -l

sudo su

b. Gain root access to machine 2 **(15 marks)**

15

Tool(s) Used: metasploit

nessus scan

Exploit(s) Used: ms03-026

C:\Windows\system32\

c. Gain root access to machine 3 **(25 marks)**

25

Tool(s) Used: msfconsole

Exploit(s) Used: unix/ftp/proftpd_133c_backdoor

set payload cmd/unix/reverse

Recreated from photographs. Highlighted: the answers written on the sheet. Boxes: the marks written by the instructor; the circled number is the total of the page.